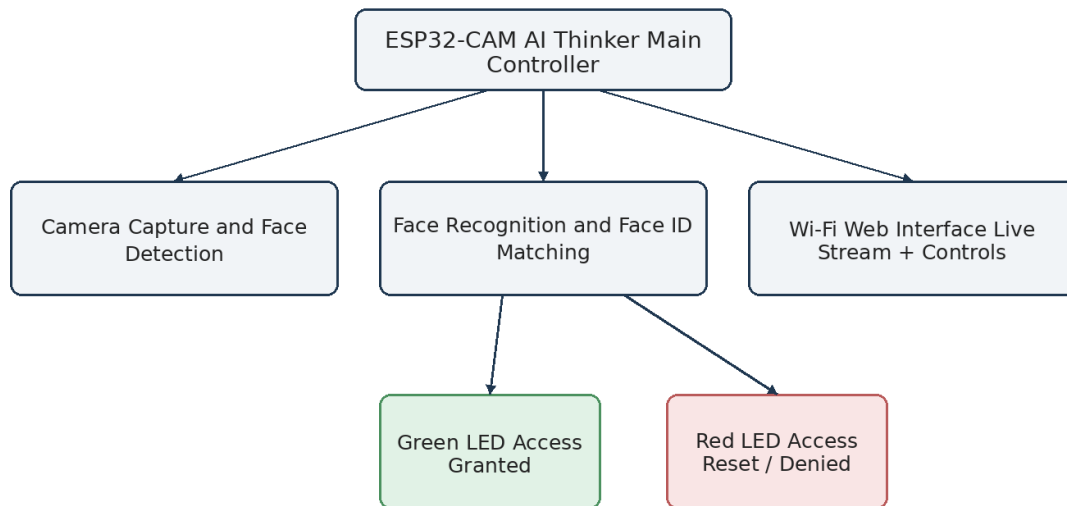


ESP32-CAM FACE RECOGNITION AND ACCESS STATUS SYSTEM

Project Documentation

System Architecture



Wi-Fi provides browser access on the same local network.

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1. Project Overview

This project is a compact ESP32-CAM prototype for face detection, face recognition, browser-based camera monitoring, and simple access-status indication. A recognized face turns the green LED on, while the red LED represents the default/reset state. The project does not use a physical door lock; LEDs are used to demonstrate the access decision.

2. Objectives

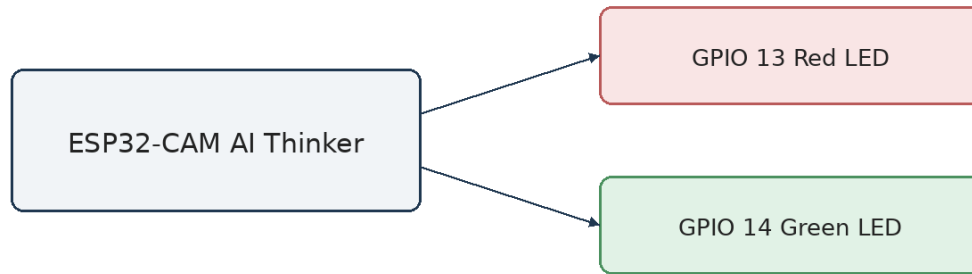
- Demonstrate face detection and face recognition on an ESP32-CAM AI Thinker board.
- Provide a browser-based interface for live camera viewing and control.
- Allow face enrollment and recognition through the embedded interface.
- Use green and red LEDs for a clear physical access-status indication.
- Build a small prototype that can be extended to a future access-control system.

3. Main Components

Component	Use
ESP32-CAM AI Thinker	Main controller, camera, Wi-Fi and face-processing platform
Camera sensor	Captures frames for detection and recognition
Green LED	Access granted indication
Red LED	Default/reset access-status indication
Breadboard and jumper wires	Prototype connections
USB / regulated 5 V supply	Power during development and testing

4. Hardware and Pin Configuration

LED Connections Used in the Prototype



Default/reset: Red ON, Green OFF

Recognized face: Green ON, Red OFF

After 6 seconds: returns to reset state

GPIO	Function	State
GPIO 13	Red LED	HIGH by default/reset; LOW when access is granted
GPIO 14	Green LED	LOW by default/reset; HIGH when access is granted
Camera GPIOs	Camera data and control	Defined by camera_pins.h for AI Thinker

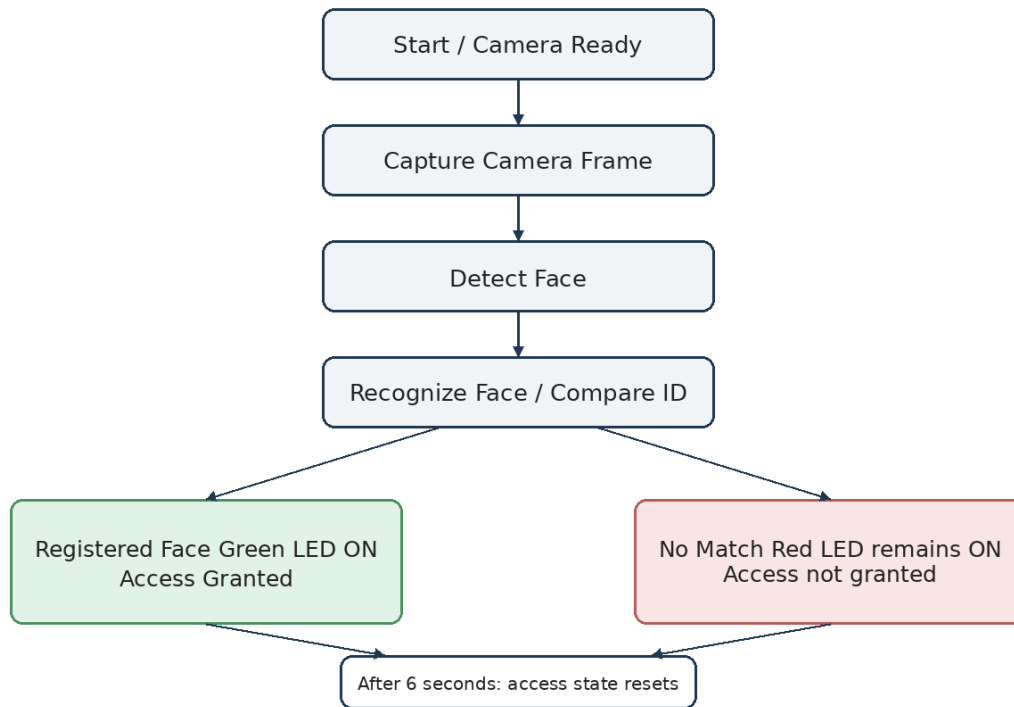
5. Software Structure

File	Purpose
face_recognition_esp32_cam.ino	Camera setup, Wi-Fi, GPIO control, server startup, access-status logic
app_httpd.cpp	Web server, streaming, capture, face detection/recognition, enrollment and control
camera_index.h	Embedded browser interface
camera_pins.h	Camera pin definitions for supported camera models

The project uses Arduino-style C/C++ code with the ESP32 camera support, Wi-Fi connectivity, and the embedded HTTP server provided by the project source.

6. Face Recognition Workflow

Recognition and Access Status Flow



7. Face Enrollment and Recognition

The web interface provides an enrollment mode for registering a face. The current implementation uses multiple samples for each face ID. During recognition, a detected face is aligned and compared with the stored face IDs. A successful match sets the matchFace state. A non-match is shown as an intruder or denied result in the camera interface and does not grant access.

8. Access-Status Logic

Condition	Green LED	Red LED	Meaning
Idle / reset	OFF	ON	Access not granted
Registered face recognized	ON	OFF	Access granted
6 seconds after grant	OFF	ON	Access state reset

The main sketch uses a 6000 ms interval. After a recognized face is detected, the green LED is activated and the timer starts. When the interval expires, the green LED is turned off, the red LED is turned on, and the temporary access state is cleared.

9. Web Interface

Endpoint	Purpose
/	Main camera web interface
/status	Camera and recognition-control status
/control	Camera and recognition control commands
/capture	Still image capture
/stream	Live MJPEG camera stream

The camera interface can be opened in a browser on the same Wi-Fi network. It provides live streaming and controls for detection, enrollment and recognition, so the prototype can be demonstrated without a separate desktop application.

10. Setup and Demonstration

- Install ESP32 board support in Arduino IDE and select the AI Thinker camera model.
- Place the four project source files in the same project structure.
- Enter the Wi-Fi SSID and password only in the private local copy of the main sketch.
- Compile and upload the sketch to the ESP32-CAM.
- Open Serial Monitor at 115200 baud and note the printed camera IP address.
- Open the IP address in a browser on the same network.
- Enable face detection and recognition, then enroll a face.
- Present the enrolled face and observe the green LED.
- Present an unregistered face and observe the denied/non-match indication.
- Wait for the six-second timer to return the LEDs to the reset state.

11. Testing

Test	Expected / Observed Behavior
Camera start-up	Camera initializes and the local web server starts
Wi-Fi connection	Board waits until the configured network is connected
Registered face	Match is detected and the green LED turns on
Unknown face	No match; access is not granted
Access timeout	After 6 seconds, green turns off and red turns on
Browser stream	Live camera view is available through /stream

12. Limitations

- The prototype uses LED feedback instead of a physical locking actuator.
- No dedicated anti-spoofing or liveness-detection mechanism is included.
- Recognition performance depends on lighting, camera position and image quality.
- The system is demonstrated on a local network rather than as a hardened internet-facing service.
- Face enrollment and stored IDs are intended for prototype use.

13. Future Improvements

- Add a buzzer for audible access feedback.
- Improve lighting or add controlled illumination for more reliable recognition.
- Add mobile or cloud notifications for access attempts.
- Add a dedicated enrollment/reset button.
- Use a stable dedicated power arrangement.
- Add event logging for recognition attempts.
- Add stronger authentication and anti-spoofing measures for any real security use.
- Add a physical lock or relay only after suitable isolation and safety design.

14. Reference

The implementation was developed with reference to the following ESP32-CAM face-recognition project, which is explicitly cited in the source sketch:

Tech StudyCell - ESP32-CAM Face Recognition Door Lock System

<https://youtu.be/PBZ-EiVIUc4>

The final prototype adapts the access indication to two LEDs: green for a recognized face and red for the default/reset state.

15. Conclusion

The project demonstrates a practical embedded computer-vision workflow using an ESP32-CAM. Camera capture, face detection, face recognition, browser-based monitoring and physical LED feedback are combined in a single prototype. The implementation provides a clear base for further development into a more complete access-control system while remaining a simple academic demonstration.